1.1 Landscape Systems — Coastal Landscapes

Systems Approach to Landscapes

System: A set of interrelated components connected by flows of energy and materials, working together as a complex whole. Coastal landscapes operate as **open systems** with inputs of energy and matter from outside the system, internal throughputs (transfers and transformations), and outputs that leave the system. The systems approach allows geographers to understand how changes to one component ripple through the entire landscape.

Systems Terminology

- **Inputs:** Energy from waves (derived from wind), tidal energy (gravitational pull of moon/sun), wind energy, solar radiation (drives weathering and evaporation); sediment from rivers, cliff erosion, offshore seabed, and longshore drift from adjacent cells; precipitation
- **Throughputs (transfers and processes):** Erosion (hydraulic action, abrasion, attrition, corrosion), weathering (freeze-thaw, carbonation, biological), mass movement (rockfall, slumping, mudflow), sediment transport (longshore drift, traction, saltation, suspension, solution), deposition
- **Outputs:** Sediment removed by longshore drift to adjacent cells or offshore, dissolved minerals lost in solution, energy dissipated as heat and sound
- **Stores:** Beaches (temporary sediment stores), sand dunes, spits, bars, mudflats, cliff faces, wave-cut platforms

Dynamic equilibrium: A balanced state in a system where inputs and outputs are roughly equal over time. The system appears stable but is constantly adjusting to small changes. Any significant change in one component (e.g. increased storm frequency, human intervention, sea-level rise) causes adjustments elsewhere to restore balance. For example, increased cliff erosion adds more sediment to the beach → wider beach absorbs more wave energy → erosion rate decreases (negative feedback). However, if the disturbance is too great, the system may shift to a new equilibrium state.

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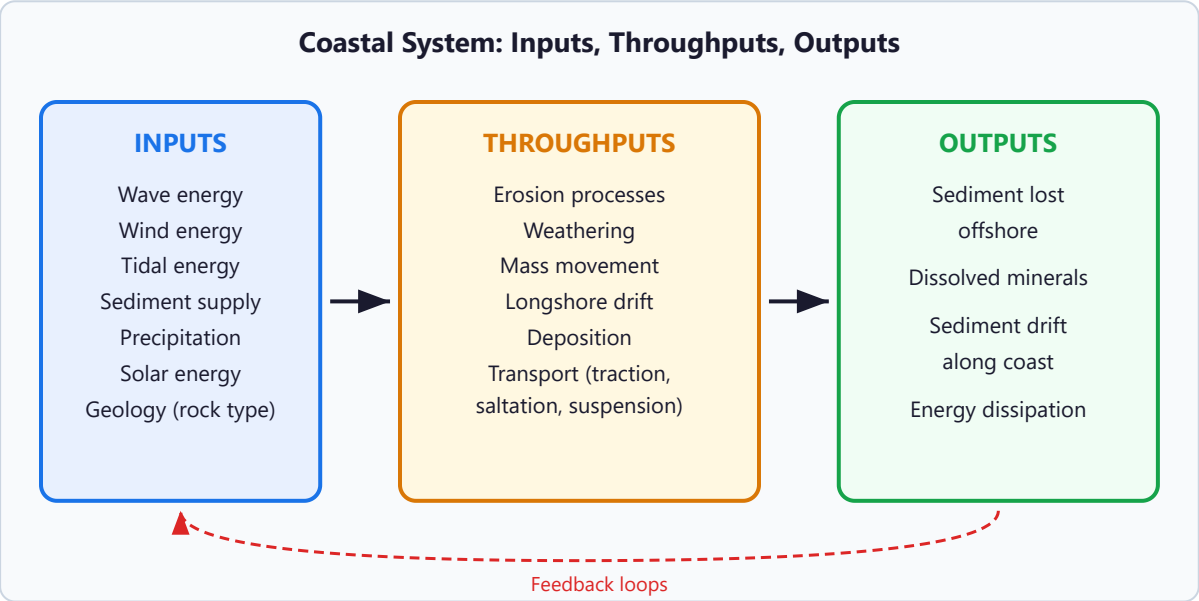
Feedback Loops

- **Negative feedback (self-regulating):** A change triggers a response that **counteracts** the original change, stabilising the system. Example: cliff collapse adds sediment to the beach → wider beach absorbs wave energy before it reaches the cliff → erosion rate decreases → less cliff collapse. This is the dominant feedback mechanism that maintains dynamic equilibrium
- **Positive feedback (self-reinforcing):** A change triggers a response that **amplifies** the original change, destabilising the system and pushing it towards a new equilibrium. Example: sea-level rise causes increased wave erosion → narrower beach → less wave energy absorption → even more erosion. Also: removal of coastal vegetation → increased erosion → less vegetation can establish → more erosion



Waves

Wave: A transfer of energy through water caused by wind blowing across the sea surface. Importantly, in open water the water itself does not travel forward — individual water particles move in circular orbits (orbital motion), returning approximately to their starting position. Only the energy moves forward. When waves approach shallow water, the circular orbits become elliptical, friction with the seabed slows the wave base, the wavelength decreases, the wave height increases, and eventually the wave breaks.

Wave Formation and Size

Three factors determine wave size and energy:

- **Wind speed:** Stronger winds transfer more energy to the water surface, generating larger, more powerful waves
- **Wind duration:** The longer the wind blows in a consistent direction, the more energy is transferred and the bigger the waves become
- **Fetch:** The distance of open water over which the wind blows without obstruction. A longer fetch allows waves to accumulate more energy. The prevailing south-westerly winds in the UK have a fetch across the entire Atlantic Ocean (~5,000 km), which is why the western coastline receives much more powerful waves than the eastern coast (short fetch across the North Sea, ~500 km)

Wave Behaviour at the Coast

In deep water, water particles move in **circular orbits** that decrease in diameter with depth. When the water depth is less than half the wavelength, the seabed interferes with these orbits (wave base touches the bottom). This causes friction, which slows the wave, steepens it, and eventually causes it to break. The **swash** is the rush of water up the beach after a wave breaks. The **backwash** is the return of water back down the beach under gravity.

Wave Refraction

When waves approach an irregular coastline with headlands and bays, the parts of the wave in shallower water near the headland slow down first, while the parts in deeper water in the bay continue at their original speed. This causes the wave fronts to **bend** (refract) and converge on the headlands, concentrating wave energy there and causing erosion. In the bays, wave energy is dispersed, favouring deposition. This is why headlands erode faster and bays tend to have beaches.

Constructive vs Destructive Waves

Feature	Constructive Waves	Destructive Waves
Frequency	Low: 6–8 waves per minute	High: 10–14 waves per minute
Wave height	Low (<1 m), long wavelength	High (>1 m), steep, short wavelength
Swash vs backwash	**Strong swash**, weak backwash — deposits more than removes	Weak swash, **strong backwash** — removes more than deposits

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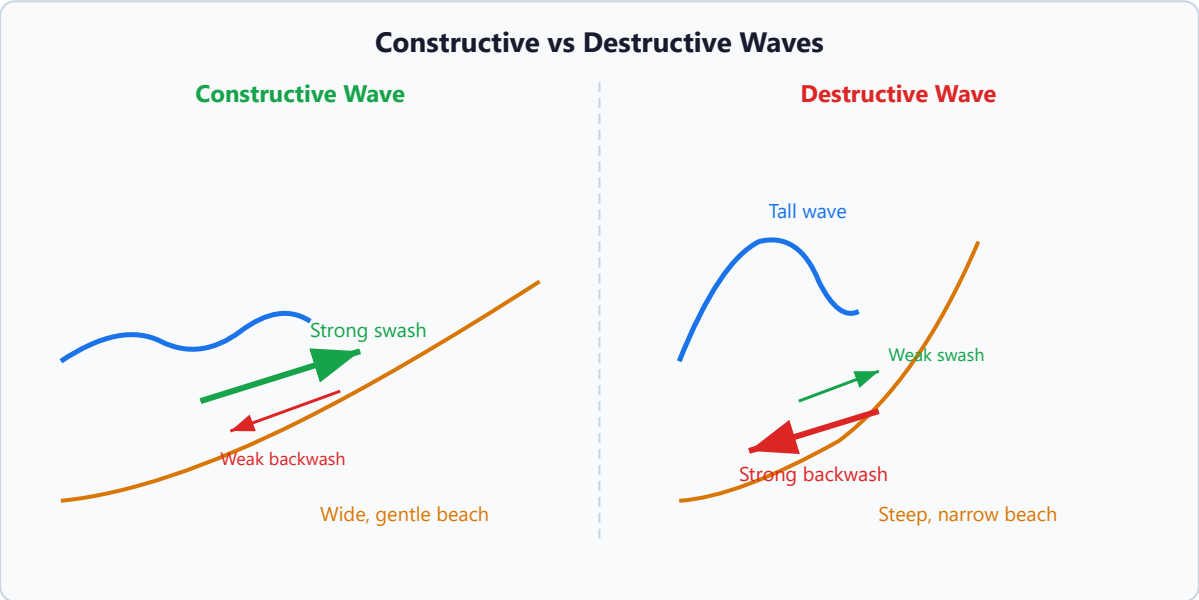
Net effect	**Build up** the beach (net deposition)	**Erode** and remove beach material
Beach profile produced	Wide, gently sloping sandy beach with berms	Steep, narrow shingle/pebble beach
Associated conditions	Calm weather; swell waves from distant storms (long wavelength)	Local storms; strong onshore winds; high wave energy
Typical environment	Sheltered bays, summer conditions	Exposed headlands, winter storms



Exam Tip: When explaining wave type, always explicitly link the swash/backwash balance to the resulting beach profile and landform. Constructive waves have a strong swash that deposits material faster than the weak backwash removes it → net deposition → wide beach. Destructive waves have a strong backwash that removes material faster than the weak swash deposits it → net erosion → narrow beach. Examiners want to see the **process-landform connection** clearly articulated.

Coastal Erosion Processes

Marine Erosion (Wave Processes)

Process	Description	Key Detail
Hydraulic action	The sheer force of water hitting the cliff face compresses air trapped in cracks, joints, and faults. The repeated compression and decompression (cavitation) gradually widens these weaknesses, breaking the rock apart.	Most effective on well-jointed rocks (e.g. chalk, limestone). During storms, wave pressure can exceed 30 tonnes per m^2
Abrasion (corrasion)	Waves pick up sand, pebbles, and rocks and hurl them against the cliff face, wearing it away like sandpaper. The eroded cliff material provides ammunition for further abrasion.	Most effective during storms when waves carry heavy sediment loads. Creates wave-cut notches at the base of cliffs. Often the most significant erosion process
Attrition	Rocks and pebbles carried by waves collide with each other and the cliff, becoming progressively smaller, smoother, and more rounded.	Explains why sediment becomes finer and more rounded downstream/downdrift. Angular boulders at cliff base → rounded pebbles → sand → silt
Corrosion (solution/chemical erosion)	Chemical dissolution of soluble rocks by slightly acidic seawater. Carbonic acid (from dissolved CO_2) reacts with calcium carbonate in limestone and chalk.	$CaCO_3 + H_2O + CO_2 \rightarrow Ca(HCO_3)_2$ (soluble). Particularly important on limestone and chalk coasts. Acts continuously, not just during storms

Sub-aerial Processes (Weathering and Mass Movement)

These operate on the cliff face from above the waterline, weakening rock and making it more vulnerable to marine erosion from below. The combination of sub-aerial weathering (top-down) and marine erosion (base) is what drives cliff retreat.

Weathering types:

- **Freeze-thaw (frost shattering):** Water enters cracks and joints during the day. At night, it freezes and expands by approximately **9%**, exerting pressure of up to 2,100 kPa on the rock. When it thaws, the water penetrates deeper into the widened crack. Repeated freeze-thaw cycles (above and below 0

degrees C) progressively shatter the rock. Most effective in environments with frequent temperature oscillations around 0 degrees C (e.g. northern UK coasts in winter)

- **Salt crystallisation:** Seawater spray enters rock pores and cracks. As water evaporates, salt crystals grow, exerting pressure on the rock and causing granular disintegration. Particularly effective on porous rocks like sandstone in warm, exposed coastal environments
- **Carbonation (chemical weathering):** Rainwater absorbs CO_2 from the atmosphere to form weak carbonic acid (H_2CO_3). This reacts with calcium carbonate in limestone and chalk, dissolving it into soluble calcium bicarbonate. Accelerated by increased atmospheric CO_2
- **Oxidation:** Iron-rich minerals in rocks react with oxygen and water, weakening the rock structure (forming rust-coloured iron oxides). Particularly visible in rocks with high iron content
- **Biological weathering:** Plant roots grow into cracks and exert pressure as they enlarge (root wedging). Burrowing organisms (piddocks, sea urchins) bore into rock, weakening it. Algae and lichens produce organic acids that chemically dissolve rock surfaces

Mass Movement

The downslope movement of rock, soil, and debris under the influence of gravity. Mass movement is triggered when the force of gravity exceeds the resisting forces (friction, cohesion) holding material in place. This balance is affected by: undercutting by wave erosion (removes support), saturation by rainfall (adds weight, reduces friction, increases pore water pressure), weathering (weakens rock), vegetation removal, and steep slope angles.

Type	Description	Speed	Conditions
Rockfall	Individual fragments or blocks of rock break away from a steep or overhanging cliff face and fall under gravity, accumulating as scree (talus) at the base	Very fast (instantaneous)	Steep, bare, well-jointed rock cliffs. Triggered by freeze-thaw weathering, undercutting, vibration
Landslide (translational slide)	A block of rock slides downslope along a relatively flat plane (bedding plane, fault surface, or clay layer). The block may remain largely intact	Fast	Layered geology with a weak layer (e.g. clay between sandstone beds). Often triggered by saturation or undercutting

Rotational slump (rotational slide)	A mass of saturated material slides along a **curved slip plane**, rotating backwards as it moves. Leaves a crescent-shaped scar at the top and a bulge of displaced material at the base	Fast to moderate	Clay or soft rock cliffs. Heavy rainfall saturates the material, adding weight and reducing shear strength. Undercutting by waves removes toe support
Mudflow	Highly saturated soil or clay flows rapidly downslope as a viscous mass	Fast	Impermeable clay-rich soils; very heavy or prolonged rainfall; steep slopes. Common on clay cliffs like Holderness
Soil creep	The very slow, continuous, downslope movement of soil under gravity	Very slow (mm/year)	Most vegetated slopes; evidenced by terracettes (small steps), tilted fence posts, curved tree trunks

Exam Tip: In exam answers about cliff retreat, always combine **marine erosion at the base** with **sub-aerial weathering and mass movement above**. The process is cyclical: (1) waves erode a notch at the base (hydraulic action + abrasion), (2) the cliff above becomes unsupported and weakened by weathering, (3) mass movement (rockfall, slump) causes the cliff to collapse, (4) debris is removed by waves, exposing a fresh cliff face, and the cycle repeats. This demonstrates the interaction of different process groups within the landscape system.

Sediment Transport

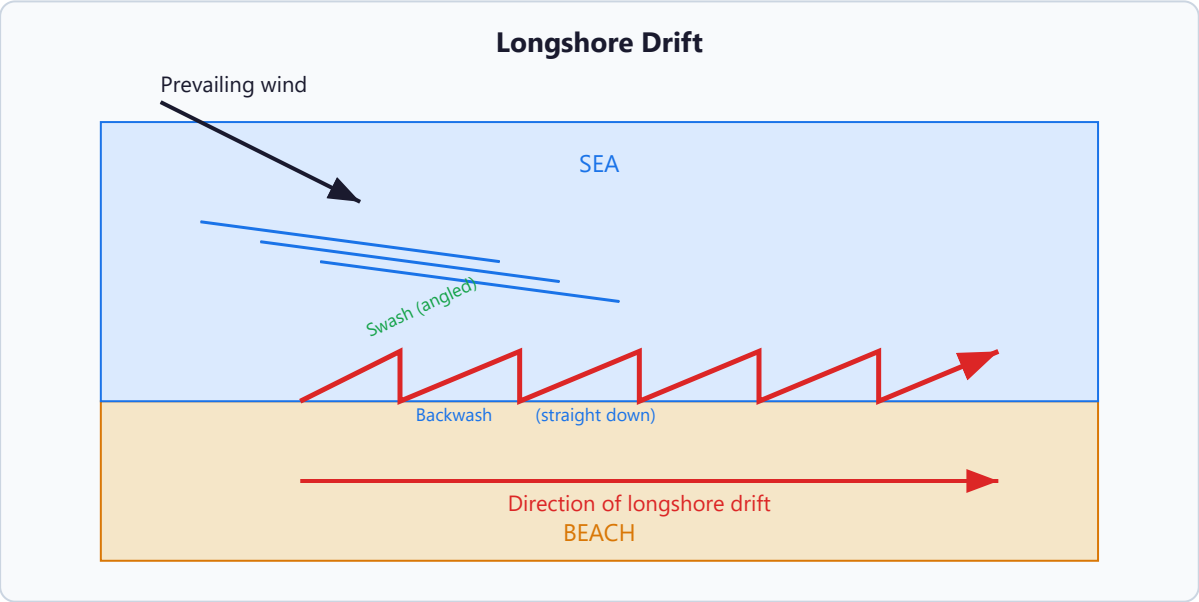
Methods of Sediment Transport

- **Traction:** Large boulders and cobbles rolled along the seabed by strong wave action. Requires the most energy
- **Saltation:** Smaller pebbles and gravel bounced along the seabed in a hopping motion by moderate wave energy

- **Suspension:** Fine sand, silt, and clay particles carried within the water column. Gives the sea a brown/turbid colour during storms. Can transport material long distances
- **Solution:** Dissolved minerals (e.g. calcium bicarbonate from limestone dissolution) carried invisibly in the water. Cannot be deposited mechanically — only through chemical precipitation or biological processes

Longshore Drift (Littoral Drift)

Longshore drift: The systematic movement of sediment along the coast by wave action. Waves approach the beach at an **angle determined by the prevailing wind direction**. The swash carries sediment up the beach at this oblique angle. The backwash, under the influence of gravity, drags sediment straight back down the beach perpendicular to the shoreline. The net result is a **zigzag movement** of sediment along the coast in the direction of the prevailing wind. On England's south coast, longshore drift generally moves sediment from west to east (prevailing south-westerly winds). On the east coast, it moves southward.



Sediment Cells (Littoral Cells)

Sediment cell: A self-contained stretch of coastline within which sediment movement (input, transport, and output) is largely independent of adjacent cells. The coast of England and Wales is divided into **11 major sediment cells**. Each cell has sources (cliff erosion, river input), transfers (longshore drift), and sinks (deposition as beaches, spits, or loss offshore). Understanding sediment

cells is crucial for coastal management because interventions in one part of a cell (e.g. building groynes) can have unintended consequences in other parts of the same cell (e.g. sediment starvation of downdrift beaches).

Coastal Deposition and Depositional Landforms

Deposition occurs when the sea loses energy and can no longer carry its sediment load. This happens when:

- Waves enter sheltered areas (bays, behind headlands, estuaries) — wave energy is reduced
- Wind speed decreases or changes direction
- There is a large sediment supply (e.g. from river mouths or rapidly eroding cliffs)
- Constructive waves dominate (strong swash deposits sediment high on the beach)
- Longshore drift reaches a change in coastline direction (material can no longer be transported forward)

Beaches

The most common depositional landform, found at the interface between land and sea. Beaches vary in composition, gradient, and morphology depending on sediment type and wave conditions:

- **Sandy beaches:** Wide, gently sloping (2–5 degrees). Fine sand is easily transported by constructive waves and deposited uniformly. Common in sheltered bays
- **Shingle/pebble beaches:** Steep (10–20 degrees). Coarse material is too heavy for the backwash to move far, so it piles up steeply. Water percolates quickly through pebbles, reducing backwash energy further (positive feedback for steepening)
- **Beach features:** Berms (ridges of material deposited at successive high tide marks), cusps (semicircular depressions formed by swash/backwash interaction), ripples (small-scale features on sandy surfaces), runnels (channels formed by drainage)

Spits

A **spit** is a long, narrow ridge of sand or shingle extending from the mainland into the sea (or across an estuary) at a point where the coastline changes direction. It forms because longshore drift continues to carry material in its original direction even though the coast turns. The end of the spit may curve landward

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(forming a **recurved tip**) due to wave refraction or secondary currents. A sheltered, low-energy area behind the spit often develops into a **salt marsh** through ecological succession (halophyte colonisation).

Case Study: Spurn Point, Humber Estuary, East Yorkshire

Spurn Point is a **5.5 km long** sand and shingle spit extending across the mouth of the Humber Estuary. It is formed by longshore drift carrying material **southward** from the rapidly eroding Holderness cliffs (the source of sediment). The spit has a pronounced recurved tip and encloses extensive mudflats and salt marshes within the estuary. Spurn is **extremely dynamic** and migrates westward over time — it has been breached and reformed at least twice in recorded history (1849 and 2013). The December 2013 storm surge breached the spit in several places, and it is now classified as a tidal island (only accessible at low tide). It is managed by the **Yorkshire Wildlife Trust** as a nature reserve and illustrates key concepts: the link between cliff erosion (source) and spit formation (store/output) within a sediment cell; the dynamic, ever-changing nature of coastal landforms; and the tension between "holding the line" and allowing natural processes.

Bars and Tombolos

- **Bar:** A spit that grows all the way across a bay, sealing off a **lagoon** of trapped water behind it. Bars form in sheltered conditions where wave energy is low enough to allow deposition across the full bay width. Example: **Slapton Ley, Devon** — a freshwater lagoon trapped behind a shingle bar
- **Tombolo:** A bar of deposited sediment that connects the mainland to an offshore island. Forms when waves refract around an island, creating a zone of constructive interference and deposition in the sheltered area between the island and the shore. Example: **Chesil Beach** connecting the Isle of Portland to the Dorset coast (28 km long, graded from pebbles at the Portland end to fine sand at the Bridport end — evidence of longshore sorting)

Sand Dunes

Formed when sand is blown inland by **onshore winds** and trapped by obstacles (driftwood, vegetation). Dunes develop through **ecological succession (psammosere)**:

1. **Embryo dunes** — nearest the strand line; small mounds around obstacles; colonised by pioneer species (sea rocket, sea sandwort) that tolerate salt spray, drought, and burial
2. **Foredunes (yellow dunes)** — taller (5–10 m); dominated by **marram grass** (*Ammophila arenaria*), which has adaptations for sand dune life: deep roots stabilise sand, rolled leaves reduce water loss, buried stems survive burial. Still very little humus in soil; alkaline pH; harsh conditions

3. **Grey dunes** — further inland; more humus develops; greater species diversity (mosses, lichens, low shrubs); more sheltered; slightly acidic soil
4. **Dune slacks** — low-lying wet areas between dune ridges where the water table is close to the surface; support wetland species (creeping willow, orchids)
5. **Mature/fixed dunes** — furthest inland; thick soil with abundant humus; climax vegetation may include shrubs, heather, and woodland; pH acidic; well-established ecosystem

Erosional Landforms

Headlands and Bays

Form where the coast is made of **alternating bands of hard and soft rock perpendicular to the shore** (a **discordant coastline**). The sea erodes the weaker rock (e.g. clay, shale) faster, forming bays. The more resistant rock (e.g. granite, limestone) protrudes as headlands. Over time, wave refraction concentrates energy on the headlands (accelerating their erosion) and disperses it in the bays (encouraging deposition of beaches). Eventually, this differential erosion will straighten the coastline — a demonstration of the system seeking dynamic equilibrium.

Cliff Profiles and Wave-Cut Platforms

1. Waves attack the base of the cliff, concentrated between **high and low tide marks** (the inter-tidal zone)
2. **Hydraulic action and abrasion** create a **wave-cut notch** (an indent at the cliff base)
3. As the notch deepens, the rock above becomes unsupported and **collapses** (mass movement)
4. The cliff retreats inland, leaving behind a gently sloping (<1–3 degrees) rocky platform — the **wave-cut platform**
5. As the platform extends seaward, waves lose energy crossing its shallow, rough surface, which gradually **slows further erosion** (negative feedback). Maximum platform width is typically ~500 m before erosion effectively ceases

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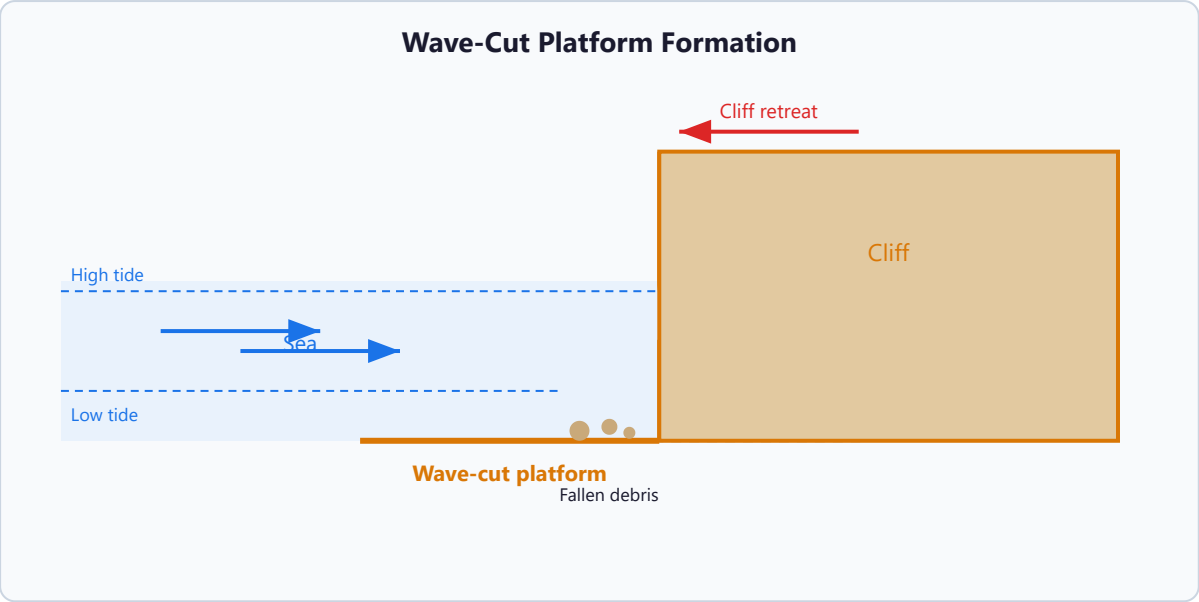

Caves, Arches, Stacks, and Stumps

A **sequence of erosional landforms** that develops on headlands, demonstrating the progressive erosion of resistant rock through geological weaknesses:

1. **Cave:** Hydraulic action and abrasion exploit a line of weakness (joint, fault, bedding plane) in the headland, eroding it into a tunnel-like cave. The back of the cave is continually eroded deeper
2. **Arch:** When caves on **both sides** of a narrow headland erode through and meet, they form an arch — a bridge of rock spanning the gap with sea on both sides
3. **Stack:** The roof of the arch is weakened by weathering (freeze-thaw, biological) and wave action from below. Eventually it **collapses**, leaving an isolated column of rock separated from the headland
4. **Stump:** Continued erosion at the base of the stack (abrasion, hydraulic action) undermines it. Eventually the stack collapses, leaving a low stump that is only visible at low tide. The stump may eventually be completely eroded away

Case Study: Old Harry Rocks, Purbeck Coast, Dorset

Located at the eastern end of the **Jurassic Coast** World Heritage Site, Old Harry Rocks are chalk stacks and stumps on the Foreland headland. **Old Harry** is a prominent stack, and **Old Harry's Wife** was a stack that collapsed to become a stump in 1896 — demonstrating the erosion sequence in action within recent history. The Foreland headland has caves, arches, and stacks in various stages of development, making it an ideal location for studying the complete erosion sequence. The chalk is relatively resistant but is heavily jointed, and these joints are exploited by hydraulic action, freeze-thaw weathering, and biological weathering (plant roots, boring organisms). The site illustrates how

geological structure (concordant/discordant coastline, rock resistance, joint patterns) controls the rate and pattern of landform development. The Jurassic Coast demonstrates coastal landforms from 185 million years of geological history in just 155 km.

Geo and Blow Hole

- **Geo:** A narrow, steep-sided coastal inlet formed when a cave roof collapses or a major joint/fault is eroded by wave action. Common on well-jointed rock coastlines
- **Blow hole:** A vertical shaft formed when wave pressure exploits a weakness in the roof of a sea cave, eventually breaking through to the surface. During storms, air and spray are forced up through the blow hole with considerable force. Example: Blow Hole, Flamborough Head, East Yorkshire

Influence of Geology on Coastal Landscapes

Rock Type and Resistance

Rock	Resistance	Typical Coastal Landforms	Why
Granite	Very resistant	Rugged headlands, towering cliffs, stacks	Crystalline, interlocking mineral structure; very hard; few weaknesses
Chalk	Resistant but permeable and jointed	White cliffs, stacks, arches (e.g. Old Harry, Seven Sisters)	Hard but joints allow hydraulic action; permeable so reduces surface runoff (less mass movement); dissolved by carbonation
Limestone (carboniferous)	Resistant but soluble and jointed	Cliffs with caves and blow holes; coastal karst features	Hard but dissolved by carbonic acid (corrosion); joints and bedding planes exploited by waves
Clay / shale	Weak, impermeable	Bays, rotational slumps, low and gentle cliff profiles, mudflows	Easily eroded; impermeable so becomes saturated after rain → mass movement; shapes bays in discordant coasts

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Sand / unconsolidated glacial till	Very weak	Rapid erosion, low crumbling cliffs (e.g. Holderness)	Loose particles easily eroded by waves and rain; no resistance to hydraulic action. Holderness boulder clay erodes at up to 2–10 m per year

Concordant and Discordant Coastlines

- **Discordant coastline:** Rock bands run **perpendicular** to the coast. Differential erosion of hard and soft rock creates alternating headlands and bays. Example: the Purbeck coast around Swanage Bay — resistant chalk forms Ballard Point headland while softer clay/sands erode to form Swanage Bay
- **Concordant coastline:** Rock bands run **parallel** to the coast. If the outer resistant rock band is breached by wave action, the softer rock behind erodes rapidly to form a cove (rounded bay). Example: **Lulworth Cove, Dorset**

Case Study: Lulworth Cove, Dorset

A textbook example of a **concordant coastline** landform. The outer band of very resistant **Portland limestone** (hard, resistant to erosion) runs parallel to the coast. At Lulworth, this limestone band was breached by wave action at a point of weakness. Behind it, the much softer **Wealden clay** was rapidly eroded into a near-circular cove (~500 m wide). The back wall of the cove is formed by more resistant **chalk**, which slows further inland erosion. Nearby **Stair Hole** shows an earlier stage in cove development: the limestone barrier is being breached in several places, and small coves are forming in the clay behind. Together, Lulworth and Stair Hole demonstrate the progressive sequence of concordant coastline evolution. The distinctive scallop shape of the cove results from wave refraction within the breached bay, which distributes erosive energy evenly around the circumference.

Coastal Management Strategies

Coastal management: Strategies used to protect coastlines from erosion and flooding, while balancing economic, social, environmental, and political considerations. The fundamental choice is between fighting natural processes (hard engineering), working with them (soft engineering), or allowing retreat (managed realignment / no active intervention).

Hard Engineering

Method	Description	Advantages	Disadvantages	Cost
Sea wall	Concrete or stone wall at the base of a cliff; often with a curved (recurved) top to deflect wave energy back seaward	Effective barrier against erosion and flooding; protects high-value property; long lifespan (50–100 years)	Very expensive; can cause wave scour at the base (undermining); creates a harsh visual impact; impedes beach access; reflected wave energy can erode the beach in front	Up to £6,000–10,000 per metre
Groynes	Wooden or stone barriers built perpendicular to the shore at regular intervals; trap sediment moved by longshore drift	Relatively cheap; effective at building up the beach (wider beach = better protection and tourism); well-understood technology	**Starve downdrift beaches of sediment** (the terminal groyne problem) — can accelerate erosion elsewhere in the sediment cell. Need regular maintenance (wood rots)	£5,000–10,000 per groyne
Rock armour (rip-rap)	Large boulders (typically granite or basalt, 5–10 tonnes each) placed at the cliff base to absorb and dissipate wave energy	Cheaper than sea walls; effective at absorbing energy; can look relatively natural; gaps allow drainage	Can be moved by severe storms; visually intrusive (especially if different rock type to local geology); difficult to walk on	£1,000–3,000 per metre
Gabions	Wire mesh cages filled with stones, placed at the cliff base	Cheap; relatively easy to construct; absorb wave energy; can be adapted to site	Short lifespan (5–10 years as wire corrodes in salt water); unattractive; individual cages can be damaged by storms	£100–500 per metre
Offshore breakwaters	Rock or concrete barriers built parallel to the coast offshore;	Reduce wave energy reaching the beach; can encourage deposition behind	Expensive; can disrupt sediment transport patterns; navigational	£3,000–5,000 per metre

	break waves before they reach the shore	them (tombolo-like features)	hazard; may alter wave patterns unpredictably	
Revetments	Sloped structures (wood, concrete, or rock) placed against the cliff base; absorb wave energy through their permeable or profiled surface	Less expensive than sea walls; can be designed to blend with environment	May need frequent replacement; less effective than sea walls against severe storms	£2,000–4,000 per metre

Soft Engineering

Method	Description	Advantages	Disadvantages	Cost
Beach nourishment (replenishment)	Adding sand or shingle to a beach, typically dredged from offshore or transported from elsewhere	Looks natural; wider beach provides better protection and attracts tourists; does not starve downdrift areas	Expensive and needs repeating every few years as storms remove material; can damage donor site ecology	£3,000–5,000 per metre (each application)
Managed retreat (managed realignment)	Allowing the sea to flood low-value land behind existing defences; often involves deliberately breaching sea walls to create new salt marshes	Cheap in the long term; creates valuable habitats (salt marshes absorb wave energy, store carbon, support biodiversity); sustainable; reduces flood risk by allowing natural floodplain function	Land and property permanently lost (compensation costs); very unpopular with affected residents and farmers; difficult political decision	Low ongoing cost (but compensation payments needed)

Dune stabilisation	Planting marram grass, installing sand fencing, creating boardwalks to reduce trampling	Cheap; natural-looking; improves biodiversity; dunes act as natural buffer against storms and flooding	Takes years to establish; vulnerable to storm damage; requires visitor management	£200–2,000 per 100 m
Cliff drainage	Installing drainage pipes or channels to remove water from cliff material, reducing saturation and pore water pressure	Effectively reduces mass movement (slumping, mudflows); relatively unobtrusive	Does not prevent marine erosion at the base; can be expensive to install and maintain; may not work on all cliff types	Variable

Shoreline Management Plans (SMPs)

SMP: A long-term strategic plan for managing flood and erosion risk along a stretch of coastline (typically one sediment cell). Developed by the Environment Agency and local authorities, they assess the economic, social, and environmental value of each section of coast and assign one of four policies for three time periods (0–20, 20–50, and 50–100 years into the future).

Policy	Meaning	Where Applied
Hold the line	Maintain or improve current coastal defences to keep the coastline in its present position	High-value settlements, infrastructure, industry (e.g. major towns, ports)
Advance the line	Build new defences seaward of the current coastline (rare and expensive)	Very high-value areas requiring new land reclamation (e.g. some ports)
Managed realignment	Allow the coastline to retreat in a controlled, planned way; may involve	Low-value agricultural land where habitat creation is beneficial; areas where

(retreat)	breaching existing defences	defending is not cost-effective
No active intervention	Allow natural processes to take their course without building or maintaining any defences	Remote, low-value, sparsely populated areas where the cost of defence exceeds the value of what is protected

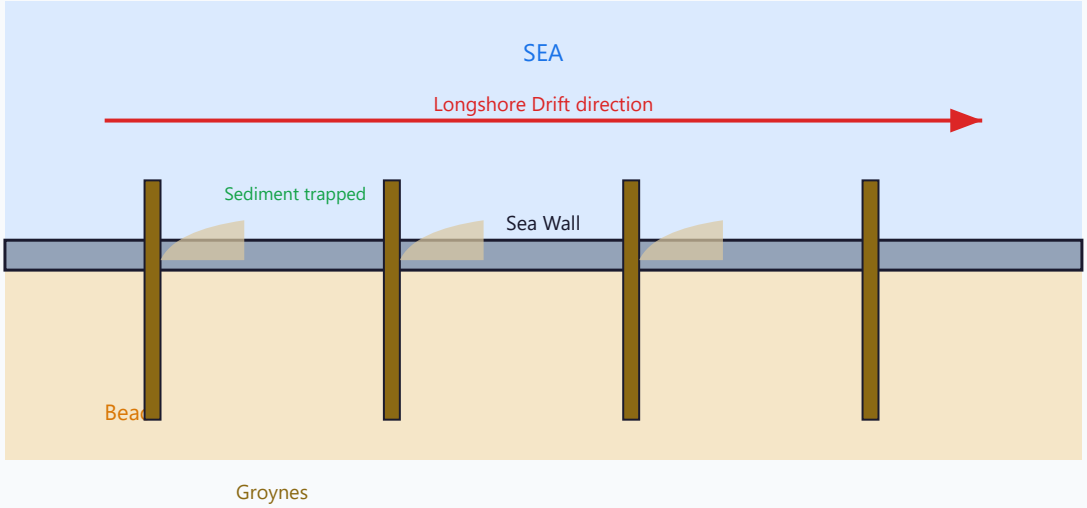
Case Study: Holderness Coast, East Yorkshire

Europe's **fastest-eroding coastline**, retreating at an average of **1.8 metres per year** (some sections up to **10 m in a single storm event**). Over 30 villages listed in the Domesday Book (1086) have been lost to the sea. The cliffs are made of soft, easily eroded **boulder clay (glacial till)** deposited during the last Ice Age — it is weakly consolidated, impermeable (so easily saturated by rain), and offers almost no resistance to wave erosion. The coast faces the North Sea with a long fetch from the north-east. Key management issues:

- **Mappleton:** Two rock groynes and granite rock armour were installed in 1991 at a cost of **£2 million** to protect the B1242 road and the village of 50 properties. They were effective locally (stabilised the cliffs). However, they **starved beaches to the south** (particularly at Cowden and Great Cowden) of sediment by intercepting longshore drift, accelerating erosion there from 1.8 m/yr to 3–4 m/yr. This is the **terminal groyne effect** — a textbook example of how protecting one section of coast can damage another within the same sediment cell.
- **Withernsea:** A sea wall and groynes protect the town centre and promenade. Tourism depends on maintaining the beach. "Hold the line" policy.
- **Skipsea:** No defences — classified as "no active intervention" in the SMP because the cost of protection exceeds the economic value of the properties at risk. Multiple homes have been lost; residents feel abandoned and argue the policy is socially unjust.
- **Spurn Point spit:** Depends on sediment eroded from Holderness cliffs being transported southward by longshore drift. Protecting the entire Holderness coast would starve Spurn of its sediment supply, potentially causing the spit to disappear — demonstrating the interconnectedness of the sediment cell system.
- **Broader lessons:** The Holderness case illustrates the conflicts between **economic value vs social justice** (who decides which communities are "worth" protecting?), **local vs strategic management** (protecting one place can harm another), and **short-term fixes vs long-term**

sustainability (hard engineering buys time but transfers problems; managed retreat is unpopular but may be the only sustainable option for much of the coast under rising sea levels).

Hard Engineering: Groynes & Sea Wall



Groynes prevent sediment loss but starve downdrift beaches of material

Climate Change and Future Challenges

Climate change poses significant challenges for coastal landscape systems:

- **Sea-level rise:** Global mean sea level is rising at ~3.6 mm per year (accelerating) due to thermal expansion of ocean water and melting of ice sheets. Higher sea levels mean waves reach further up the cliff, increasing erosion rates and flooding frequency. The IPCC projects 0.3–1.1 m rise by 2100 depending on emissions scenario
- **Increased storminess:** Climate models suggest more frequent and intense storms in the North Atlantic, increasing destructive wave energy and the frequency of storm surge events. The 2013/14 winter storms caused severe coastal damage across southern England
- **Changes to sediment supply:** Altered wave patterns and storm tracks may change longshore drift patterns, disrupting established sediment cells and the landforms they sustain
- **Implications for management:** Many current defences were designed for historical conditions and may be inadequate for future sea levels and storm intensities. The economic cost of "holding the line" everywhere is unsustainable. Managed retreat may become the default policy for many stretches of

coast, raising difficult social and political questions about which communities to protect and which to abandon

Putting It All Together: The Coastal System

Exam Tip: In 20-mark essays, use the **systems framework** to structure your answer. Show how inputs (wave energy, geology, sediment supply), throughputs (erosion, transport, deposition), and outputs (sediment loss, landform creation) interact to create the landscape. Always include **specific place examples** with data (e.g. Holderness erodes at 1.8 m/yr; Mappleton groynes cost £2 million). Use **correct terminology** (dynamic equilibrium, negative/positive feedback, sediment cell, concordant/discordant). Link management strategies back to the system — explain how human intervention disrupts dynamic equilibrium and creates unintended consequences elsewhere in the system. For the highest marks, discuss how landscape systems are being altered by **climate change** and evaluate whether current management approaches are sustainable in the long term.

Key System Connections

- Cliff erosion (throughput) provides sediment input → transported by longshore drift → deposited as spits, beaches, bars (stores)
- Geology (input) determines cliff resistance → controls rate and pattern of erosion → shapes headlands, bays, and cliff profiles
- Wave type (constructive/destructive) determines whether the coastline is net eroding or net accreting
- Human management interrupts natural processes → unintended consequences in other parts of the sediment cell (e.g. terminal groyne effect)
- Climate change alters inputs (more energy, higher sea level) → positive feedback → accelerated change → management challenges
- The coastal landscape is always evolving — it is a dynamic system, not a static one

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